

Ibrahim Reda Wasfy

Automotive & Design Engineer | Fresh Graduate

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Obour City ,Cairo, Egypt | Military Service Exempted | Egyptian | Driving License

PROFILE

Automotive engineering graduate from **Helwan University - Mataria** with a lifelong passion for motorsport that has shaped every professional choice I've made. With hands-on leadership experience in two competitive teams: serving as **Head of Vehicle Dynamics** in Helwan Racing Team (E-Rally), and currently **Team Leader** in HFS Racing Team. Throughout both roles, I have taken ownership of the full vehicle development cycle — spanning engineering calculations, CAD design, simulation, manufacturing, assembly, and testing — while leading subteams, coordinating workflows. I am a detail-oriented, motivated engineer eager to bring that competition-proven engineering mindset to a full-time role in automotive development.

EDUCATION

Bachelor Of Automotive Engineering,

[Capital \(Helwan\) University - Mataria Branch](#)

09/2021 – 06/2026

Cairo, Egypt

Relevant Coursework: [Vehicle Technology](#), [Vehicle Performance](#), [Vehicle Dynamics](#), [Automotive Fault Diagnosis](#), [Automotive Maintenance and Repair](#), [Stress Analysis](#), [Mechanical Design](#), [Computer Aided Design](#), [Automotive Chassis Design](#), [Mechatronics](#), [Automotive Electrical and Electronic Systems](#), [Automotive Control Systems Design](#).

Secondary School, [El Safwa Language School](#)

09/2014 – 06/2021

Obour, Cairo, Egypt

EXPERIENCE

Helwan Formula Student (HFS) Racing Team

[Team Leader](#)

06/2025 – Present

Cairo, EGY

- Re-established the Helwan Formula Student Racing Team after an 11-year absence.
- Led a multidisciplinary team of 10 engineers across vehicle dynamics, powertrain, aerodynamics, and chassis to develop a Formula Student for FSUK competition.
- Managed project timeline and milestone planning to meet FSUK deadlines.
- Represented the team in external communications with sponsors and industry partners, supporting recruitment and funding efforts.
- Coordinated all administrative processes for international competition entry.

[Vehicle Dynamics](#)

09/2025 – Present

Cairo, EGY

- Directed vehicle dynamics development encompassing steering geometry design and Ackermann optimization, simulating steering block geometry in SolidWorks to validate turn-in response and low-speed maneuverability achieving a turning radius of ~4.5 m.
- Designed a rack and pinion steering system from scratch, including steering arm integration into the upright, and validate geometry and packaging.
- Utilized Optimum Kinematics to optimize toe arm angle and minimize bump steer, ensuring steering precision and stability across the full suspension travel range.

Lap Time Simulation (LTS) Engineer

04/2026 – 06/2026

Cairo, EGY

- Developed structured test plans using Excel tools for test management, data collection, result comparison, and performance tracking and simulation matrices across iterative CarMaker LTS simulation runs

Helwan Racing Team (E-Rally)

Head of Vehicle Dynamics

03/2025 – 01/2026

Cairo, EGY

- Led the Vehicle Dynamics department, setting technical direction, managing budget and timelines, and overseeing manufacturing execution through to assembly.
- Implemented design revisions based on prior-year performance data to drive iterative development.
- Mentored junior engineers in vehicle dynamics fundamentals, ensuring knowledge continuity across the department.

Vehicle Dynamics Vice Head

09/2024 – 03/2025

Cairo, EGY

- Mechanical design, calculations, and engineering BOM & DBR for Urban and Prototype, building both from scratch.
- Oversaw the full manufacturing lifecycle from technical drawings through production, assembly, and testing, ensuring design specifications were maintained end-to-end
- Prepared and onboarded new candidates into the vehicle dynamics department

Vehicle Dynamics Member

07/2023 – 09/2024

Cairo, EGY

- Performed full steering and brake system calculations for a 256 kg urban EV, defining geometry and pedal ratios to meet EVER competition technical requirements.
- Mechanical design and FEA for steering parts.
- Authored the Vehicle Dynamics Design Report for EVER competition submission.

Revitup Drifting Competition , *Motorsport Organizer*

12/2022 – 09/2024

Cairo, EGY

- Set up competition track layout on-site
- Enforced safety protocols for drivers, vehicles, spectators, and facilities, managing on-site incident response and damage assessment

PROJECTS

Formula Student One-Seater Racing Car , *FSUK*

07/2025 – 07/2026

- Owned the complete steering system development from concept to final design, defining geometry based on Ackermann principles and tire slip characteristics to optimize handling performance and cornering response, through geometry analysis.
- Developed and validated a full rack and pinion steering system from scratch, encompassing ratio selection, rack travel, steering effort analysis, and steering box assembly integration with chassis and suspension constraints
- Conducted multiple design iterations to balance steering effort, responsiveness, and packaging limitations
- Contributed to technical documentation and design justification aligned with FSUK engineering judging criteria
- *Tools Used: Microsoft Excel - SolidWorks - MATLAB - Simulink - Optimum Kinematics*

Gasoline Urban Car , *Shell Eco-marathon Qatar*

07/2024 – 10/2025

- Designed, manufactured, and tested from concept to competition, following a full (Calculation - Design - Manufacturing - Testing) for participating at SEM 2025
- *Tools Used: Microsoft Excel - SolidWorks*

Electric Prototype Car [🔗](#), *Shell Eco-marathon Qatar* 07/2024 – 10/2025

- Designed, manufactured, assembled, and tested in parallel with the Gasoline Urban Car for Shell Eco-Marathon competition entry
- *Tools Used: Microsoft Excel - SolidWorks*

Electric Urban Car [🔗](#), *EVER Egypt* 07/2023 – 07/2024

- Built from scratch passing by different stages mainly: (Calculation - Design - Manufacturing - Assembly - Testing), to participate at EVER Egypt
- *Tools Used: Microsoft Excel - SolidWorks*

INTERNSHIPS

Ezz-Elarab Automotive Group [🔗](#), *Engineer Intern* 09/2024

- Completed automotive engineering training covering vehicle systems, technical coursework, and new model induction sessions for the all new Jeep Grand Cherokee. 6th of Oct, Al Jizah, Egy

MCV - Manufacturing Commercial Vehicles [🔗](#), *Workshop Engineer Intern* 08/2024

- Gained hands-on exposure to commercial vehicle manufacturing across Buses, Trucks, and Vans departments, observing engine overhaul operations El Obour, Al Qalyubiyah, Egy

ESOTICA Service Centre [🔗](#), *Workshop Engineer Intern* 07/2024

- Assisted in diagnosing and resolving a timing system fault on a McLaren 570S, supporting root cause analysis and corrective repair
- Supported workshop service operations for high-performance vehicles Industrial zone, New Cairo, Egy

MTI Jaguar Land Rover Egypt [🔗](#), *Workshop Engineer Intern* 06/2024

- Completed Pre-Delivery Inspection (PDI) procedures and test plan documentation for Range Rover and Jaguar model lines El Obour, Al Qalyubiyah, Egy

Abou Ghali Automotive [🔗](#), *Workshop Engineer Intern* 09/2023

- Performed diesel engine servicing and supported workshop operations, assisting the workshop manager with daily job-card scheduling and technician coordination El Obour, Al Qalyubiyah, Egy

Elsaba Auto-Service Nissan [🔗](#), *Workshop Engineer Intern* 07/2023

- Conducted scheduled maintenance services and systematic visual inspections across a 10+ vehicle daily operation following Nissan OEM service protocols katameya, Cairo, Egypt

ACHIEVEMENTS

Shell - Eco marathon [🔗](#), *Doha, Qatar* 01/2026

- 6th Place in Urban Gasoline Category (2026)
- 15th Place in Electric Prototype Category (2026)

EVER Egypt [🔗](#), *North Coast, Egypt* 08/2024

- 1st Place Overall Dynamic Track
- 1st Place Design
- 1st Place Cost

ORGANISATIONS

Automobile and Touring Club Egypt (ATCE) *🔗, Time Keeper*

04/2026 – Present

Cairo, Egypt

- Operated official FIA timing systems to record and validate lap times across competitive categories and coordinate with race control to ensure accurate results and regulatory compliance throughout event sessions.

COURSES AND CERTIFICATIONS

SolidWorks: CSWA, CSWP (in progress) • **Matlab:** Fundamentals • **FIA:** Soft Skills Development •

HP Academy: Motorsport Fabrication Fundamentals, Wiring Fundamentals

TECHNICAL SKILLS

CAD | DFM | FEA | SolidWorks | MATLAB | Simulink | AutoCAD | GD&T | Vehicle Dynamics | Optimum Kinematics | Car Maker | Technical Report | BOM | Root Cause Analysis | Excel Sheets

SOFT SKILLS

Team leadership | Team Work | Multi-tasking | Negotiation | Project Management | Report writing

LANGUAGES

Arabic: (Native) | **English:** (Full professional proficiency) | **French:** (Beginner) | **German:** (Beginner)